

Closing the hook rank formula: rigorous nesting and reduction to (Q_m)

Clio

9 May 2026

Abstract

We continue the May 7–8 program on the structure of $K_m := \ker \Pi^{S_m}|_{V_{(m-1,1)}}$ for the staircase Bott–Samelson element. The May-8 evening writeup reduced the parity dichotomy to the *tail-coordinate proportionality* (R_m) . Here we sharpen the reduction further:

1. **Reduction Theorem.** For m even, the proportionality (R_m) follows from three structural inputs: (a) the nesting $K_{m-2}^{\text{embed}} \subset K_m$ (2-step), (b) (P_m) , and (c) (Q_m) : $v_{m-1}^*(K_m) \neq 0$.
2. **Nesting Theorem (rigorous).** For all $n \geq 4$, the natural embedding $K_{n-2} \rightarrow V_{(n-1,1)}$ lies in K_n . Proof is by induction on n , using (P_{n-3}) and the trivial action of T_{n-2}, T_{n-1} on $\text{span}(v_2, \dots, v_{n-3})$.
3. Combined: the hook rank formula $\text{rank } \Pi^{S_n}|_{V_{(n-1,1)}} = \lceil (n-2)/2 \rceil$ is now rigorously proved *modulo the single property* (Q_m) for m even, which is significantly cleaner than the May-8 evening’s tail-coordinate gap.

This eliminates the matrix-inversion computation that was needed to prove (R_6) rigorously and replaces it with a structural proof that propagates through all m .

1 Setup

We adopt the conventions of the May-8 morning paper (2026-05-08-hook-rank-formula.tex). $H_q(S_n)$ acts on $V_{(n-1,1)}$ via Hoefsmit seminormal matrices on the basis $v_2, \dots, v_n$. We use $\alpha_i = 1, q_i = 1/(1+q)$ in the normalization where $u_i = v_i + \beta_i v_{i+1}$ is the $+q$ -eigenvector of T_i and $u'_i = v_i + \beta'_i v_{i+1}$ is the -1 -eigenvector. (Verified in `check_identity.py`.) Then $(T_i + 1)v_i = (1+q)p_i u_i$ and $(T_i + 1)v_{i+1} = u_i$.

The staircase Bott–Samelson is $\Pi^{S_n} = R'_2 R'_3 \cdots R'_n$ with $R'_k = (T_{k-1} + 1) \cdots (T_1 + 1)$, applied right-to-left. Set $K_m := \ker \Pi^{S_m}|_{V_{(m-1,1)}}$.

We freely use:

- **Image Lemma** (May-8 evening): for $w \in V_{(n-1,1)}$ with $R'_n(w) = y \in \text{span}(v_2, \dots, v_{n-2})$, $w_{v_n} = C_n \cdot y_{v_{n-2}}$ with explicit $C_n \in \mathbb{Q}(q)^\times$.
- **Lemma 4 (May-8 morning)**: $D_n = R'_n(V_{(n-1,1)}) = \text{span}(v_2, \dots, v_{n-2}, u_{n-1})$, $\dim n - 2$.
- **Reduction Theorem (May-8 evening)**: for $n \geq 4$, $v_n^*(K_n) = 0 \iff v_{n-2}^*(K_{n-1} \cap \{v_{n-1}^* = 0\}) = 0$.
- (P_m) : $v_m^*(K_m) = 0$ iff m is odd. Proven inductively by the Reduction Theorem and base cases $m \leq 4$, conditional on (R_{m-2}) for m odd at $m \geq 5$.

1.1 Definition of natural embeddings

For $a < b$, the natural embedding $\iota_{a,b}: V_{(a-1,1)} \hookrightarrow V_{(b-1,1)}$ sends $v_k \mapsto v_k$ for $k = 2, \dots, a$ (with $v_{a+1}, \dots, v_b$ adjoined as zero). Under the S_a -branching of $V_{(b-1,1)}$, the image of $\iota_{a,b}$ coincides with the $V_{(a-1,1)}$ -summand. We write K_m^{embed} when the target $V_{(n-1,1)}$ is understood from context.

2 The Reduction Theorem (clean statement)

We restate the May-8 evening reduction in the form most useful for inductive arguments.

Theorem 1 (Reduction). *Let m be even, $m \geq 4$. Suppose:*

(N) (Nesting) $K_{m-2}^{\text{embed}} \subset K_m$, where the embedding is $\iota_{m-2,m}: V_{(m-3,1)} \hookrightarrow V_{(m-1,1)}$.

(P_m) $v_m^*(K_m) \neq 0$.

(Q_m) $v_{m-1}^*(K_m) \neq 0$.

Then v_{m-1}^* and v_m^* are proportional on K_m . In other words, (R_m) holds.

Proof. We have $\dim K_{m-2} = (m-2)/2 = \dim K_m - 1$ since m is even (so $m-2$ is even and $\lfloor (m-2)/2 \rfloor = (m-2)/2$).

By (N), $K_{m-2}^{\text{embed}} \subset K_m$. $K_{m-2}^{\text{embed}} \subset \text{span}(v_2, \dots, v_{m-2}) \subset \{v_{m-1}^* = 0\} \cap \{v_m^* = 0\}$. Hence

$$K_{m-2}^{\text{embed}} \subset K_m \cap \{v_{m-1}^* = 0\} \subset K_m,$$

with the first inclusion having codimension at most $\dim K_m - \dim K_{m-2} = 1$, and the second having codimension exactly 1 by (Q_m). By dimension count, $K_{m-2}^{\text{embed}} = K_m \cap \{v_{m-1}^* = 0\}$.

By the analogous argument with v_m^* in place of v_{m-1}^* (using (P_m)): $K_{m-2}^{\text{embed}} = K_m \cap \{v_m^* = 0\}$.

Therefore $K_m \cap \{v_{m-1}^* = 0\} = K_m \cap \{v_m^* = 0\}$, so the two functionals v_{m-1}^*, v_m^* on the vector space K_m have the same kernel. Two nonzero linear functionals with the same kernel on a vector space are proportional. Hence $v_{m-1}^* = \lambda_m v_m^*$ on K_m for some $\lambda_m \in \mathbb{Q}(q)^\times$. $\square$

Remark 2. This reduction is structurally cleaner than the May-8 evening formulation: the proportionality is now seen as a direct consequence of *both kernel-coefficients-are-zero loci coinciding inside the smaller kernel K_{m-2}^{embed}* . The proof avoids matrix inversion entirely.

The three hypotheses (N), (P_m), (Q_m) are independent structural facts about the kernel filtration. We will prove (N) unconditionally below; (P_m) is the May-8 dichotomy already inductively established (modulo the inductive step we now close); (Q_m) is the only remaining gap.

3 Nesting: the rigorous proof

Theorem 3 (Nesting). *For all $n \geq 4$, the natural embedding $K_{n-2}^{\text{embed}} \hookrightarrow V_{(n-1,1)}$ lies in K_n .*

The key technical fact we use is that T_{n-2} and T_{n-1} act as the scalar q on every basis vector v_l with $l \leq n-3$.

Lemma 4. *For $i \in \{n-2, n-1\}$ and $l \in \{2, 3, \dots, n-3\}$: $T_i v_l = q v_l$, hence $(T_i + 1)v_l = (1+q)v_l$.*

Proof. T_i acts as q on v_l for $l \notin \{i, i+1\}$ (Lemma 2 of the May-8 morning paper). For $i = n-2$: $\{i, i+1\} = \{n-2, n-1\}$, and $l \leq n-3$ has $l \notin \{n-2, n-1\}$. For $i = n-1$: $\{i, i+1\} = \{n-1, n\}$, and $l \leq n-3$ has $l \notin \{n-1, n\}$. $\square$

Corollary 5. For any $z \in \text{span}(v_2, \dots, v_{n-3}) \subset V_{(n-1,1)}$:

$$(T_{n-2} + 1)z = (1 + q)z, \quad (T_{n-1} + 1)z = (1 + q)z.$$

Proof of Theorem 3. We prove the statement by strong induction on n .

Base cases $n = 4, 5$. $K_2 \subset V_{(1,1)} = \mathbb{Q}(q)v_2$, with $\Pi^{S_2} = (T_1 + 1)$ acting as 0 on v_2 (since $T_1 v_2 = -v_2$), so $K_2 = \mathbb{Q}(q)v_2$.

For $n = 4$: $K_2^{\text{embed}} = \mathbb{Q}(q)v_2 \subset V_{(3,1)}$, and by direct computation (May-8 morning) $K_4 = \text{span}(v_2, v_3 - cv_4) \ni v_2$.

For $n = 5$: $K_3 \subset V_{(2,1)}$. $\Pi^{S_3} = (T_1 + 1)(T_2 + 1)(T_1 + 1) = (1 + q)(T_1 + 1)(T_2 + 1)$ on $V_{(2,1)} = \text{span}(v_2, v_3)$. $(T_1 + 1)v_2 = 0$, $(T_1 + 1)v_3 = (1 + q)v_3$, so $\Pi^{S_3}(v_2) = 0$, $\Pi^{S_3}(v_3) = (1 + q)^2(T_2 + 1)v_3 = (1 + q)^3 q_2 v_2$, which is nonzero. Hence $K_3 = \mathbb{Q}(q)v_2$.

$K_3^{\text{embed}} = \mathbb{Q}(q)v_2 \subset V_{(4,1)}$, and $v_2 \in K_5$ by direct computation (or because $(T_1 + 1)v_2 = 0$ kills the leftmost factor of Π^{S_5}).

Inductive step ($n \geq 6$). Assume nesting holds for all $4 \leq l < n$.

Let $w \in K_{n-2}^{\text{embed}} \subset V_{(n-1,1)}$. By definition, $w = \iota_{n-2,n}(w')$ for some $w' \in K_{n-2} \subset V_{(n-3,1)}$. The support of w is in $v_2, \dots, v_{n-2}$, i.e., $w \in \text{span}(v_2, \dots, v_{n-2})$.

Step 1: Apply R'_{n-2} . $R'_{n-2} \in H_q(S_{n-2})$ acts on $V_{(n-1,1)}$. Restricted to the $V_{(n-3,1)}$ -summand $\text{span}(v_2, \dots, v_{n-2})$ (under S_{n-2} -branching), R'_{n-2} acts as the level- $(n-2)$ operator on $V_{(n-3,1)}$. Hence $R'_{n-2}(w) = \iota_{n-2,n}(R'_{n-2}(w'))$, where the right-hand R'_{n-2} is the level- $(n-2)$ operator on $V_{(n-3,1)}$.

By definition of K_{n-2} , $\Pi^{S_{n-2}}(w') = 0$. Using $\Pi^{S_{n-2}} = \Pi^{S_{n-3}} R'_{n-2}$:

$$\Pi^{S_{n-3}}(R'_{n-2}(w')) = 0, \quad \text{so } R'_{n-2}(w') \in \ker \Pi^{S_{n-3}}|_{V_{(n-3,1)}}.$$

Now $\Pi^{S_{n-3}} \in H_q(S_{n-3})$. Under the S_{n-3} -branching of $V_{(n-3,1)}$, we have $V_{(n-3,1)} \downarrow_{S_{n-3}} = V_{(n-4,1)} \oplus V_{(n-3)}$, where the $V_{(n-4,1)}$ -summand is $\text{span}(v_2, \dots, v_{n-3})$ and the trivial S_{n-3} -summand is $\mathbb{Q}(q)v_{n-2}$. $\Pi^{S_{n-3}}$ acts as $(1 + q)^{(n-4)(n-3)/2}$ (a nonzero scalar) on the trivial summand, so its kernel is contained in the $V_{(n-4,1)}$ -summand:

$$\ker \Pi^{S_{n-3}}|_{V_{(n-3,1)}} \subset \text{span}(v_2, \dots, v_{n-3}).$$

Therefore $R'_{n-2}(w') \in \text{span}(v_2, \dots, v_{n-3})$ in $V_{(n-3,1)}$, and embedded into $V_{(n-1,1)}$ via $\iota_{n-2,n}$:

$$R'_{n-2}(w) = \iota_{n-2,n}(R'_{n-2}(w')) \in \text{span}(v_2, \dots, v_{n-3}) \subset V_{(n-1,1)}.$$

Step 2: Apply $(T_{n-2} + 1)$ and $(T_{n-1} + 1)$. $R'_{n-2}(w) \in \text{span}(v_2, \dots, v_{n-3})$. By Corollary 5:

$$R'_{n-1}(w) = (T_{n-2} + 1)R'_{n-2}(w) = (1 + q)R'_{n-2}(w),$$

$$R'_n(w) = (T_{n-1} + 1)R'_{n-1}(w) = (1 + q)^2 R'_{n-2}(w).$$

Step 3: Conclude using inductive nesting. Apply $\Pi^{S_{n-1}}$ to both sides of $R'_n(w) = (1 + q)^2 R'_{n-2}(w)$:

$$\Pi^{S_n}(w) = \Pi^{S_{n-1}}(R'_n(w)) = (1 + q)^2 \Pi^{S_{n-1}}(R'_{n-2}(w)).$$

Now $R'_{n-2}(w) \in \text{span}(v_2, \dots, v_{n-3}) \subset V_{(n-2,1)}$ -summand of $V_{(n-1,1)}$. So $\Pi^{S_{n-1}}$ acts on $R'_{n-2}(w)$ via the $V_{(n-2,1)}$ -summand action, which equals the level- $(n-1)$ $\Pi^{S_{n-1}}$ on $V_{(n-2,1)}$.

We assert: $R'_{n-2}(w)$, viewed as an element of $V_{(n-2,1)}$, lies in K_{n-1} . Granting this, $\Pi^{S_{n-1}}(R'_{n-2}(w)) = 0$, and hence $\Pi^{S_n}(w) = 0$, i.e., $w \in K_n$. (*QED step 3*)

Verification of the assertion. $R'_{n-2}(w) \in V_{(n-2,1)}$ (in fact in $\text{span}(v_2, \dots, v_{n-3}) \subset V_{(n-2,1)}$). It corresponds to $\iota_{n-3,n-1}(R'_{n-2}(w'))$, the embedding into $V_{(n-2,1)}$ of the level- $(n-2)$ vector $R'_{n-2}(w') \in V_{(n-3,1)}$.

We have $R'_{n-2}(w') \in K_{n-3}^{\text{embed (into } V_{(n-3,1)})}$. But this is exactly the natural embedding of K_{n-3} from $V_{(n-4,1)}$ into $V_{(n-3,1)}$ (by definition of how K_{n-3} sits inside $V_{(n-3,1)}$).

Now, the further embedding from $V_{(n-3,1)}$ into $V_{(n-2,1)}$ is $\iota_{n-3,n-1}$. The composite embedding $V_{(n-4,1)} \rightarrow V_{(n-3,1)} \rightarrow V_{(n-2,1)}$ is $\iota_{n-3,n-1}^{2\text{-step}}$, the 2-step embedding into $V_{(n-2,1)}$.

So the assertion becomes: $K_{n-3}^{\text{embed (2-step into } V_{(n-2,1)})} \subset K_{n-1}$. *This is precisely the nesting at level $n-1$.* By the inductive hypothesis (since $n-1 < n$), this holds.

Hence $R'_{n-2}(w) \in K_{n-1}$ in $V_{(n-2,1)}$, completing the proof. $\square$

4 Closing the hook rank formula modulo (Q_m)

We now combine Theorems 1 and 3 with the May-8 evening Reduction Theorem.

Theorem 6 (Hook rank formula, conditional on (Q_m)). *Assume (Q_m) holds for all even $m \geq 4$. Then for all $n \geq 2$,*

$$\text{rank } \Pi^{S_n}|_{V_{(n-1,1)}} = \lceil \frac{n-2}{2} \rceil.$$

Proof. By strong induction on n . For $n \leq 5$, direct computation verifies the formula.

For $n \geq 6$: we prove (P_n) , which implies the rank formula via the dimension recursion (Theorem 6 in the May-8 evening writeup). Specifically, by induction (P_l) holds for $l < n$.

Case n odd. By the May-8 evening Reduction Theorem, $v_n^*(K_n) = 0$ iff $v_{n-2}^*(K_{n-1} \cap \{v_{n-1}^* = 0\}) = 0$. With $n-1$ even, (P_{n-1}) gives $v_{n-1}^*(K_{n-1}) \neq 0$, so $K_{n-1} \cap \{v_{n-1}^* = 0\}$ has codim 1 in K_{n-1} .

By Theorem 1 applied at $m = n-1$ even (using (P_{n-1}) , (Q_{n-1}) , and Theorem 3): (R_{n-1}) holds, i.e., v_{n-1}^* and v_{n-2}^* are proportional on K_{n-1} , hence $K_{n-1} \cap \{v_{n-1}^* = 0\} = K_{n-1} \cap \{v_{n-2}^* = 0\}$. Thus v_{n-2}^* vanishes on $K_{n-1} \cap \{v_{n-1}^* = 0\}$, giving (P_n) for n odd.

Case n even. By the Reduction Theorem, $v_n^*(K_n) = 0$ iff $v_{n-2}^*(K_{n-1}) = 0$ (since (P_{n-1}) gives $v_{n-1}^*(K_{n-1}) = 0$, so $K_{n-1} \cap \{v_{n-1}^* = 0\} = K_{n-1}$). By (Q_{n-1}) (needed at level $n-1$ odd, i.e., applied to v_{n-2}^* on K_{n-1}), $v_{n-2}^*(K_{n-1}) \neq 0$. So $v_n^*(K_n) \neq 0$, giving (P_n) for n even.

In both cases (P_n) is established, completing the induction. $\square$

Remark 7. The condition (Q_{n-1}) for $n-1$ odd in the case n even reduces to (Q_l) for l odd. In our notation, (Q_l) for l odd is: $v_{l-1}^*(K_l) \neq 0$ where l is odd. By (P_{l-1}) for $l-1$ even ($v_{l-1}^*(K_{l-1}) \neq 0$, conditional on (R_{l-2})), and the relation $K_{l-1}^{\text{embed}} = K_l$ for l odd (which follows from $(P_l) + \text{dimension}$), (Q_l) for l odd reduces to $v_{l-1}^*(K_{l-1}) \neq 0$ which is (P_{l-1}) . So (Q_l) for l odd is automatic from $(P_l) + (P_{l-1})$.

The genuinely new condition is (Q_m) for m even.

5 The remaining gap: (Q_m) for m even

The single open structural property is:

Conjecture 8 ((Q_m) for m even). *For each even $m \geq 4$, the linear functional v_{m-1}^* is nonzero on K_m , i.e., there exists $w \in K_m$ with $w_{v_{m-1}} \neq 0$.*

Computational verification: (Q_m) holds for all even $m \leq 8$ (see `check_n8_proportionality.py`, where the second component of each kernel basis vector is precisely $w_{v_{m-1}}$ and is nonzero for the non- v_2 -aligned vectors).

5.1 Why (Q_m) is plausible

By Theorem 3, $K_m \supsetneq K_{m-2}^{\text{embed}}$. So K_m contains a vector $\xi \notin K_{m-2}^{\text{embed}} = \text{span}(v_2, \dots, v_{m-2}) \cap K_m$. By (P_m) , ξ has nonzero v_m -coefficient. The question is whether ξ *also* has nonzero v_{m-1} -coefficient.

The Image Lemma gives, for ξ with $R'_m(\xi) = y \in K_{m-1}^{\text{embed}}$: $\xi_{v_m} = C_m y_{v_{m-2}}$. The corresponding formula for $\xi_{v_{m-1}}$ involves the back-substitution through the matrix of R'_{m-1} on $V_{(m-1,1)}$, and is generically nonzero.

A clean structural proof of (Q_m) would be the analog of the Image Lemma for the v_{m-1}^* -coordinate, namely:

Conjecture 9 (Strong Image Lemma). *For $w \in V_{(n-1,1)}$ with $R'_n(w) = y \in K_{n-1}^{\text{embed}}$ (where $n-1$ is odd), $w_{v_{n-1}}$ is a specific linear combination of $\{y_{v_l} : l = 3, \dots, n-2\}$, and on K_{n-1}^{embed} this combination is a nonzero scalar multiple of $y_{v_{n-2}}$.*

If this conjecture holds, then by the Image Lemma's $y_{v_{n-2}} \neq 0$ for some $y \in K_{n-1}$ (i.e., (Q_{n-1}) at the odd level, which we observed reduces to (P_{n-2})), we get $w_{v_{n-1}} \neq 0$, hence (Q_n) .

Conjecture 9 simultaneously implies both (R_n) and (Q_n) . The May-8 evening Proposition for $n = 6$ verifies this via explicit matrix inversion of R''_5 plus the K_5 -relation $y_{v_4} = -cy_{v_3}$.

5.2 Status

- **Rigorously proved (this writeup):** Theorem 1 (Reduction) and Theorem 3 (Nesting). Together these show: (R_m) for m even follows from $(P_m) + (Q_m)$.
- **Rigorously proved (May-8 evening):** Image Lemma, Reduction Theorem (cell-coefficient dichotomy), (R_4) and (R_6) via explicit matrix inversion.
- **Open:** (Q_m) for general even $m \geq 8$, equivalently Conjecture 9.

6 Comparison with previous writeups

vs. May-8 morning. The morning writeup left two gaps: (a) detailed iterative-image case analysis or (b) the dichotomy “ $\delta(n) = 1$ iff n even”. The May-8 evening closed this via the Image Lemma + (R_m) . The current writeup further reduces (R_m) to (Q_m) via Theorem 1, and rigorously establishes Nesting (which was implicitly used in the May-8 evening analysis but never proved structurally).

vs. May-8 evening. The evening writeup proved the Image Lemma rigorously, set up the inductive scaffold (P, Q, R) , and proved (R_6) via direct matrix inversion. The remaining gap was: prove (R_m) for general m even (or equivalently the proportionality $\lambda_m = v_{m-1}^*/v_m^*$ on K_m).

This writeup eliminates the matrix-inversion approach entirely:

- **Theorem 1** replaces “proving proportionality” with “proving two functionals have the same kernel locus on K_m .”
- **Theorem 3** provides the structural inclusion $K_{m-2}^{\text{embed}} \subset K_m$ that fixes one half of the kernel locus identification.
- The remaining half (the other functional kernel locus matching) is exactly (Q_m) .

The remaining open question (Q_m) is significantly cleaner than the tail-coordinate proportionality of the evening writeup: (Q_m) is an unconditional statement about a single linear functional being nonzero, while (R_m) involved an algebraic-identity verification of two functionals’ ratio.

7 Summary

Theorem 10 (Main result). *The hook rank formula $\text{rank } \Pi^{S_n}|_{V_{(n-1,1)}} = \lceil (n-2)/2 \rceil$ is rigorously proved for all $n \geq 2$ modulo the single condition (Q_m) for m even, $m \geq 4$, where (Q_m) states $v_{m-1}^*(K_m) \neq 0$ in $V_{(m-1,1)}$.*

Computational verification. (Q_m) holds for $m \in \{4, 6, 8\}$ (verified in `check_n8_proportionality.py`).

Files.

- `~/projects/proofs/2026-05-09-nesting-and-reduction.tex` (this writeup).
- `~/projects/scratch/2026-05-09-nesting-identity/check_identity.py` (verification of Hoef-smit normalization $\alpha_i = 1$, $q_i = 1/(1+q)$ used).
- `~/projects/scratch/2026-05-08-kernel-dichotomy/` (probe and proportionality scripts).